

ATHARVA GAUR

atharvagaur7487@gmail.com | +91-8128927559 | [GitHub](#) | [LinkedIn](#) | [LeetCode](#)

Education

B.Tech — Computer Science Engineering (AI & ML) | VIT Vellore | Aug 2024 – May 2028

CGPA: 9.53/10 • Coursework: DSA, OOP, OS, DBMS, Computer Networks, Linear Algebra, Discrete Mathematics

Class XII — Science (CBSE, 92.4%) | Air Force School, Subroto Park | 2024

TECHNICAL SKILLS

- Languages: C++, Python, C, JavaScript
- Software Engineering: Object-Oriented Programming, Software Architecture, System Design, Design Patterns, Git, Linux, Debugging, Testing, Performance Optimization
- Data Structures & Algorithms: Arrays, Linked Lists, Trees, Graphs, Dynamic Programming, Greedy, Binary Search, BFS, DFS
- AI / ML: RAG, LangChain, FAISS, TensorFlow/Keras, Scikit-learn, XGBoost, LightGBM, Feature Engineering, Model Evaluation, SHAP
- Cloud & Infrastructure: Amazon Web Services (AWS), Amazon EC2, Amazon S3, Basic Cloud Computing
- Frameworks & Tools: NumPy, Pandas, Git/GitHub
- Computer Science Fundamentals: Data Structures, Algorithms, Operating Systems, Database Management Systems, Computer Networks, Concurrency, Multi-threading, Distributed Systems, Client-Server Architecture

PROJECTS

RAG-Based Intelligent Learning Assistant | [LIVE DEMO](#) [GITHUB](#)

- Architected a modular RAG system using Lang Chain, FAISS and LLMs for document-based question answering.
- Designed semantic chunking, vector retrieval and prompt pipelines for scalable knowledge retrieval.
- Improved retrieval quality through embedding optimization and modular software design.
- Developed a web-based document question-answering application integrating backend retrieval pipelines with an interactive user interface.

Merchant-Based Fraud Detection System | **ML Research Project**

- Architected a modular big-data analytics pipeline processing over 7M financial transactions.
- Automated feature engineering and evaluation workflows using XGBoost, LightGBM and Random Forest.
- Engineered 25+ predictive features while eliminating temporal data leakage through time-aware validation.
- Improved model reliability using SHAP and ablation studies, achieving 94%+ F1-score and PR-AUC of 0.91 across ensemble models.

Neural Language Prediction System | **Deep Learning Project** | [GITHUB](#)

- Architected an embedding + stacked LSTM sequence model for next-word prediction, achieving 75–80% accuracy.
- Evaluated model performance using perplexity and accuracy metrics; tuned via dropout regularization

COMPETITIVE PROGRAMMING/CODING PROFILES

- Solved 200+ algorithmic programming problems across LeetCode, CodeChef and Codeforces using C++.
- Strong understanding of graph algorithms, dynamic programming, binary search, greedy algorithms and tree-based data structures.

Languages: English (Fluent), Hindi (Native), German (Basic)